

Package #161

incorporate a multiphase nanocomposite oxygen■scavenging additive, analogous to the MNCP, into the pouch■cell electrolyte or separator to actively oxidize residual oxygen during formation cycling, thereby preventing gas evolution

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'gas-induced swelling' concerning lithium-ion pouch cell. User-submitted engineering problem (session input): "Why do lithium-ion pouch cells blister and swell during formation cycling before any external abuse, and how can we suppress gas-induced swelling without adding a vent?". ROUTING: nhtsa_complaints: not queried — no vehicle (make|model|year) extracted from the problem; this source answers vehicle-parameterized questions only. [nhtsa_complaints] retrieval status NOT_QUERIED_GRAMMAR (no vehicle extracted; source grammar is make|model|year (routing decision, not absence)) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] 10 records retrieved live (2026-2026). Illustrative records: 'Understanding the thermal stability of high-energy density alloy anodes is critical for the safe deployment of sodium-ion batteries (SIBs). Here, accelerating rate calorimetry (ARC), post-mortem chara'; 'Formation of robust solid/cathode electrolyte interphases (S/CEI) is vital for long-term stability and high-performance operation of lithium-ion batteries (LIBs), particularly under high voltage regim'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: oxygen oxidation (as a chemical reaction that consumes oxygen and does not generate gas)

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.